Week 3 –Exploratory Data Analysis: Data Preparation

COMP517 Data Analysis

SCHOOL OF ENGINEERING, COMPUTER AND MATHEMATICAL SCIENCES

This Week

- Recap of Previous Lecture
- Exploratory Data Analysis
- Data Analysis: Summarizing Variables
  - Measures of Location or Central Tendency
  - Measures of Spread
  - Frequency Distribution
- Data Cleaning and Preparation
  - What is Noisy Data?
  - Handling Missing Data

Data Types (organization and format)

Data can be classified into structured, semi-structured, and unstructured data.

Structured Data

Highly organized and easily searchable by simple, straightforward algorithms. This type of data is typically stored in relational databases and spreadsheets.

- ✓ Tabular Format: often organized into tables with rows and columns
- ✓ Schema: predefined schema that defines the data types, constraints, and relationships between different elements in the dataset.
- ✓ High Integrity: Structured data usually enforces data integrity constraints, ensuring that data meets certain rules and standards.

Characteristics of Structured Data

- Dimensionality
- Sparsity
- Resolution
- Distribution

  Refers to how values are spread or arranged across a dataset.

  ✓ Uniform Distribution
    - ✓ All values are equally likely to occur within a given range (e.g. Rolling a fair die)

  ✓ Normal Distribution, also known as Gaussian distribution
    - ✓ where data tends to cluster around a central value with no bias left or right. (e.g. Heights of adults in a population.)

  ✓ Skewed Distribution
    - ✓ Distribution where data is not symmetric and tends to have a long tail on one side. (e.g. income distribution)

Quantitative Attributes

- Involve values that come in meaningful (not arbitrary) numbers
  - Examples: age, weight annual salary
  - The numbers can be discrete or continuous

1. Discrete Attributes: have a countable number of possible values
   - ✓ Limited and distinct values
   - ✓ For example, the on a die come in discrete values (1, 2, 3, 4, 5, or 6)
2. Continuous Attributes: represent data that can take on any value within a specific range.
   - usually numeric and can be measured with precision
   - typically represent real numbers
   - Examples: height of a person, the weight of an object, or the temperature of a location.

Categorical (or Qualitative) Attributes

- Attributes whose values fall into some category, indicating a quality or property of an object
- Examples :Gender, ethnicity, product group
- Generally expressed in text strings, but not always; sometimes coded using numerical values:
  - Gender: 0 – male; 1 – female
  - Ethnicity: 0 – Caucasian; 1 – Maori; 2 – Pacific Islander

1. Nominal Attribute – Variables without any natural order such as the examples above
2. Ordinal Attribute – Variables whose categories can be put into some natural order such as:

   Product satisfaction ranking “Not satisfied”……….”Extremely satisfied”

Data Wrangling

- **Data Cleaning:** Identifying and handling missing, incorrect, or inconsistent data values. This involves imputing missing data, correcting errors, and resolving inconsistencies to ensure data quality.
- **Data Transformation:** Converting data from one format to another or reshaping data to fit the desired structure. This may involve aggregating, summarizing, or normalizing data for easier analysis.
- **Data Integration:** Combining data from multiple sources or databases into a unified dataset for analysis. This step ensures that all relevant information is available in a single, coherent format.
- **Data Reduction:** Reducing the amount of data while preserving its informational content. This is often done to improve processing efficiency and reduce computational resources.
- **Data Formatting:** Ensuring that data is presented in a consistent and standardized manner. This may involve converting data types, handling units, or applying formatting rules.

Data Quality

- How can we **detect** problems with the data?
- What can we do about these problems?

1. Data Profiling: Understand the structure, content, and relationships within your data
   - Analyze distributions, identify missing values (NULLs), check for outliers, and assess data types.

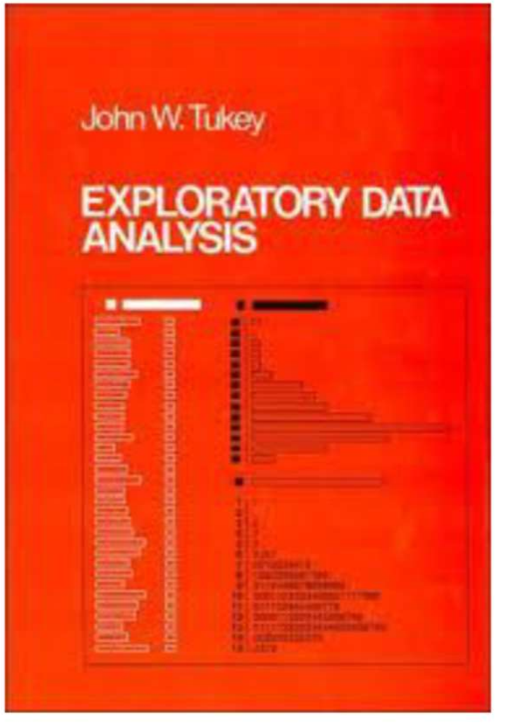
     E.g., Look for columns with a high number of NULL values or identify unexpected ranges for numerical data
2. Data Validation:
   - Verify that data conforms to predefined rules and constraints

     E.g., Ensure dates are in the correct format, numeric values are within acceptable ranges

- Examples of data quality problems:
  - noise and outliers
  - missing values
  - duplicate data

Exploratory Data Analysis 1977

- Based on insights developed at Bell Labs in the 60's
- Techniques for visualizing and summarizing data
- What can the data tell us? (in contrast to "confirmatory" data analysis)
- Introduced many basic techniques:
  - 5-number summary, box plots, stem and leaf diagrams,...
- 5 Number summary:
  - extremes (min and max)
  - median & quartiles
  - More robust to skewed & longtailed distributions



A nice online introduction can be found in Chapter 1 of the NIST Engineering Statistics Handbook
http://www.itl.nist.gov/div898/handbook/index.htm

WHAT IS Exploratory Data Analysis (EDA)?

- The analysis of datasets based on various numerical methods and graphical tools.
- Exploring data for patterns, trends, underlying structure, deviations from the trend, anomalies and strange structures.
- It facilitates discovering unexpected as well as conforming the expected.
- Another definition: An approach/philosophy for data analysis that employs a variety of techniques (mostly graphical).

AIM OF THE EDA

- Maximize insight into a dataset
- Uncover underlying structure
- Extract important variables
- Detect outliers and anomalies
- Test underlying assumptions
- Develop valid models

AIM OF THE EDA

- The goal of EDA is to open-mindedly explore data.
- Tukey: *EDA is detective work… Unless detective finds the clues, judge or jury has nothing to consider.*
  - Here, judge or jury is the confirmatory data analysis
- Tukey: *Confirmatory data analysis goes further, assessing the strengths of the evidence.*
- With EDA, we can examine data and try to understand the meaning of variables.
  - *E.g.*, What do the abbreviations stand for?

Exploratory vs Confirmatory Data Analysis

EDA	CDA
• No hypothesis at first	• Start with hypothesis
• Generate hypothesis	• Test the null hypothesis
• Uses graphical methods (mostly)	• Uses statistical models

STEPS OF EDA

- Generate good research questions
- Data restructuring: You may need to make new variables from the existing ones.
  - Instead of using two variables, obtaining rates or percentages of them
  - Creating dummy variables for categorical variables
- Based on the research questions, use appropriate graphical tools and obtain descriptive statistics
- Try to understand the data structure, relationships, anomalies, unexpected behaviors
- Try to identify confounding variables, interaction relations and multicollinearity, if any
- Handle missing observations
- Decide on the need of transformation
- Decide on the hypothesis based on your research questions

AFTER EDA

Confirmatory Data Analysis:

- Verify the hypothesis by statistical analysis
- Draw conclusions
- Present your results

Classification of EDA*

- EDA is generally cross-classified in two ways.

1. Each method is either non-graphical or graphical.
2. Each method is either univariate or multivariate (usually just bivariate).
   - Univariate methods look at one variable (data column) at a time
   - Bivariate EDA looks at exactly two variables.
   - Multivariate methods look at two or more variables at a time to explore relationships.

- Non-graphical methods generally involve calculation of summary statistics, while
- Graphical methods obviously summarize the data in a diagram or pictorial.
- It is almost always a good idea to perform univariate EDA on each of the components of a multivariate EDA before performing the multivariate EDA.

*Seltman, H.J. (2018). Experimental Design and Analysis.
http://www.stat.cmu.edu/~hseltman/309/Book/Book.pdf

EXAMPLE 1

Data from the Places Rated Almanac (Boyer and Savageau, 1985)

9 variables from 329 metropolitan areas in the USA

1. Climate mildness
2. Housing cost
3. Health care and environment
4. Crime
5. Transportation supply
6. Educational opportunities and effort
7. Arts and culture facilities
8. Recreational opportunities
9. Personal economic outlook

+ latitude and longitude of each city

Questions:

1. How is climate related to location?
2. Are there clusters in the data (excluding location)?
3. Are nearby cities similar?
4. Any relation between economic outlook and crime?
5. What else???

EXAMPLE 2

In a breast cancer research, main questions of interest might be:

- Does any treatment method result in a higher survival rate?
- Can a particular treatment be suggested to a woman with specific characteristic?
- Is there any difference between patients in terms of survival rates (*E.g.,* Are white women more likely to survive compared to black women at the same stage of disease?)

EXAMPLE 3

In a project, investigating the well-being of teenagers after an economic hardship, main questions can be:

- Is there a positive (and significant) effect of economic problems on distress?

- Which other factors can be most related to the distress of teenagers? *E.g.,* age, gender,…?

EDA Part 2: Summarizing Data With Tables and Plots

Examine the entire data set using basic techniques before starting a formal statistical analysis.

- Familiarize yourself with the data
- Find possible errors and anomalies
- Generate summary statistics; Examine the distribution of values for each variable

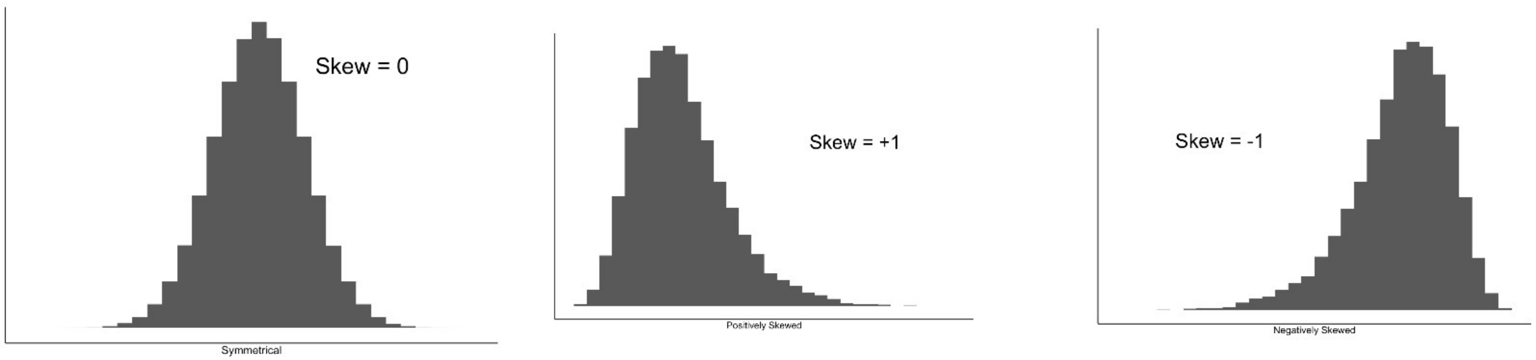
Summary (or Descriptive) Statistics

Measures of Central Tendency or Location

- It is useful to define numerical measures that describe important features of the data
- One important way is by use of an *average*
- Any measure indicating the center of a set of data, arranged in order of magnitude, is called a *measure of location* or a *measure of central tendency*
- The most used such measures are the mean, median and mode.
  - Most summary statistics can be calculated in a single pass through the data

Summary Statistics: Descriptive Statistics

- For quantitative variables (and possibly for ordinal variables) it is worth while looking at the central tendency:
  - spread, skewness, and kurtosis of the data for a particular variable from an experiment.



Symmetrical = skew close to 0 (i.e. in range -0.5 to 0.5)

Percentiles

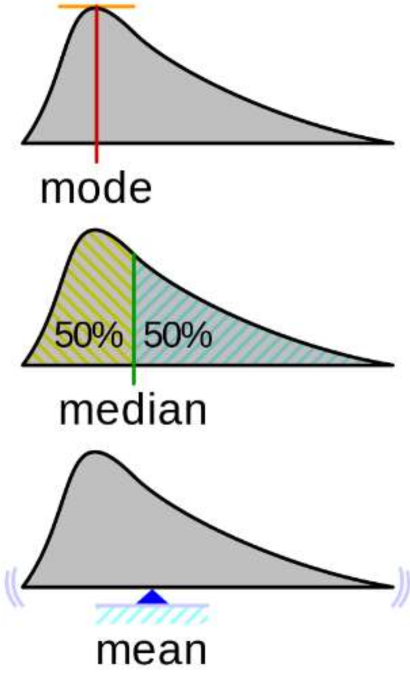
- For continuous data, the notion of a percentile is more useful. A percentile is a measure that shows how a value or a score compares to other values or scores in a group of data.

- It indicates the percentage of data that is equal to or below that value or score.

- For instance, the 20^{th} percentile is the value (or score) below which 20% of the observations may be found.

Measures of Location: Mean and Median

- The mean is the most common measure of the location of a set of points.
- However, the mean is very sensitive to outliers.
- Thus, the median or a trimmed mean is also commonly used.

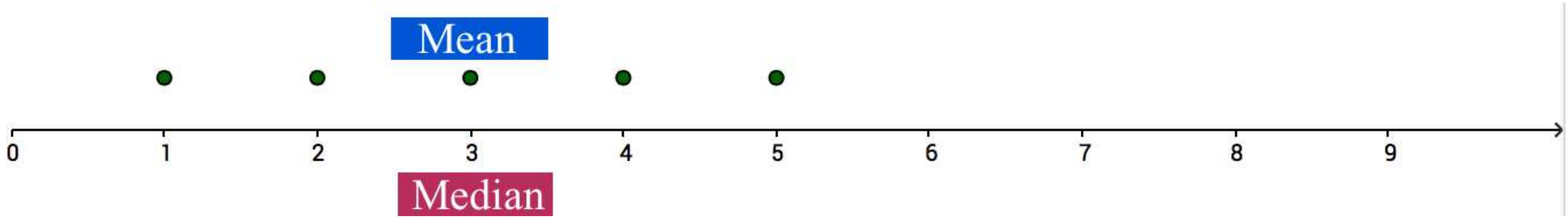


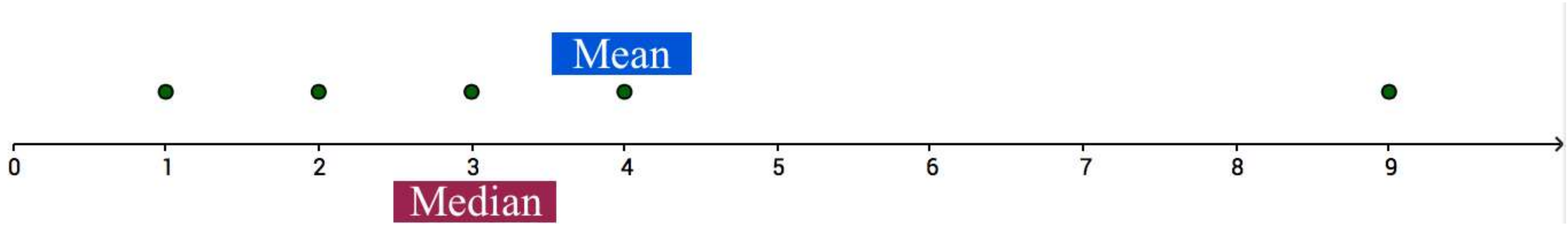
$$\text{mean}(x) = \overline{x} = \frac{1}{m}\sum_{i=1}^{m} x_i$$

$$\text{median}(x) = \begin{cases} x_{(r+1)} & \text{if } m \text{ is odd, i.e., } m = 2r+1 \\ \frac{1}{2}(x_{(r)} + x_{(r+1)}) & \text{if } m \text{ is even, i.e., } m = 2r \end{cases}$$

Centrality

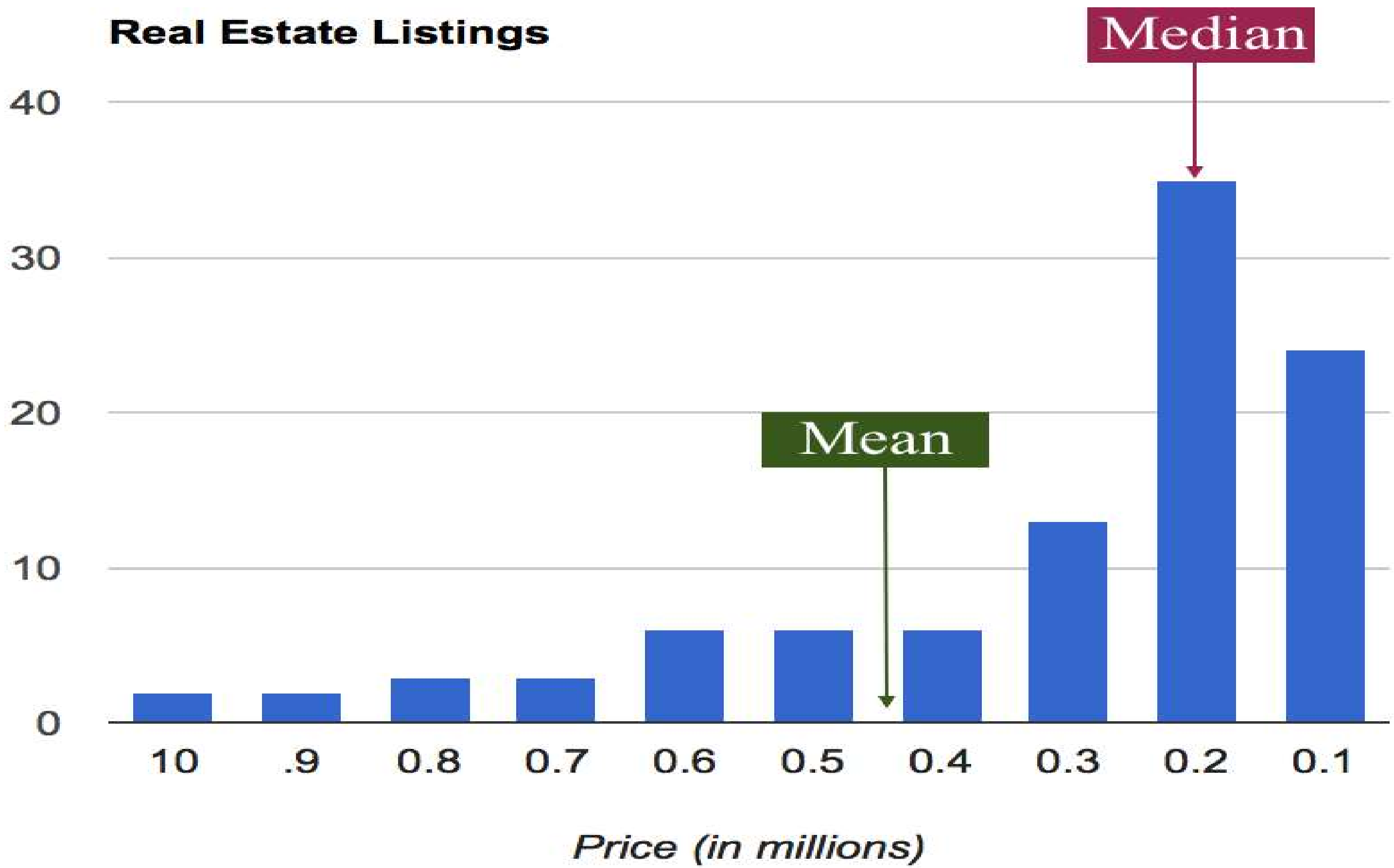
The mean is *sensitive to outliers*.





Centrality

The mean is sensitive to *skewness (asymmetry)* of *distributions*.



Measures of Spread: Range

- The spread of samples measures how well the mean or median describes the sample set.
- We need to know how the observations spread out from the average
- Example: Orange Juice bottled by 2 companies in ml.s

Charlie's	501	502	501	503	503
Freshly	509	504	494	493	510

- Both samples have the same mean of 502 ml.s
- The *variation* or *dispersion* of the observations from the average is less for Charlie's than for Freshly

Measures of Spread: Range

- The most important measures of variation are the *Range*, *Variance*, and *Standard deviation*
- Range is the difference between the largest and smallest values

$$\text{Range} = \text{Maximum Value} - \text{Minimum Value}$$

- In the orange juice companies' example, the range for Charlie's is 2 compared to a range of 17 for Freshly
- Thus, the range is also sensitive to outliers,
  - as it considers only the extreme values
  - And tells us nothing about the distribution of values in between
- so other measures are often used.

Measures of Spread: Variance

- The (sample) variance, denoted s^2, measures how much on average the sample values "deviate" from the mean
- The variance (or standard deviation) is the most common measure of the spread of a set of points.

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

n is the total number of data points in the sample.

- The (sample) standard deviation, denoted s, is the square root of the variance.

$$s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

Statistical Description of Data

- Frequency and Mode
- It is often useful to group data into classes (bins) and determine the number of values that fall in each class
  - Frequency Distribution
- The frequency of an attribute value is the percentage of time the value occurs in the data set
  - For example, given the attribute 'gender' and a representative population of people, the gender 'female' occurs about 50% of the time.
- The mode of an attribute is the most frequent attribute value
- The notions of frequency and mode are typically used with categorical data

Summarizing Variables

The goal for both categorical and continuous data is data reduction while preserving/extracting key information about the process under investigation.

- Categorical variables
  - Frequency tables - how many observations in each category?
  - Relative frequency table - percent in each category
  - Bar chart and other plots
- Continuous variables
- Bin the observations (create categories .e.g., (0-10), (11-20), etc.) then, treat as ordered categorical
  - Plots specific to Continuous variables

Categorical Data Summaries

Example:

New Cancer Cases in the U.S. for Selected Sites, 2001

Colon	Breast	Prostate	Lung	Urinary
135,400	193,700	198,100	169,500	87,500

Cancer site is a variable taking 5 values

- categorical or continuous?
- ordered or unordered?

Frequency Table

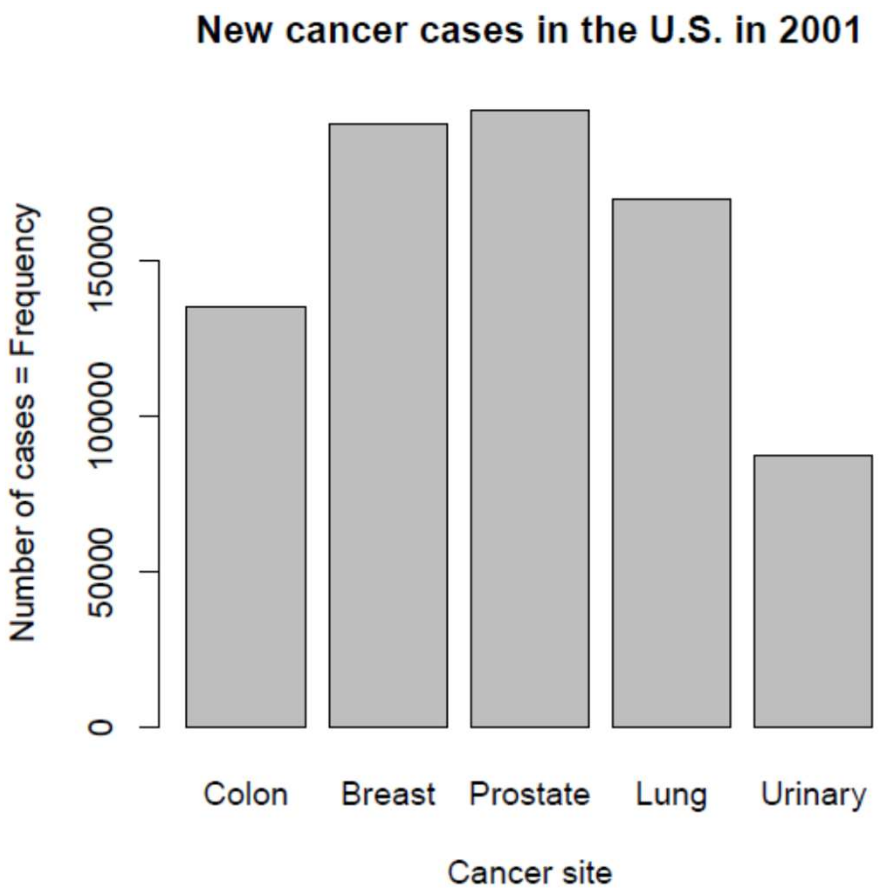
Cancer by Site, 2001

	Colon	Breast	Prostate	Lung	Urinary	Total
Freq.	135,400	193,700	198,100	169,500	87,500	**784,200**
Relative Freq.	17.26%	24.70%	25.26%	21.61%	11.16%	100%

- Frequency Table: Categories with counts
- Relative Frequency Table: Percentage in each category

Graphing a Frequency Table - Bar Chart:

- The information in a frequency distribution is easier to grasp if presented graphically
- *E.g.,* Plot the number of observations in each category:



Continuous Data - Tables

Example: Ages of 10 adult leukemia patients:

35; 40; 52; 27; 31; 42; 43; 28; 50; 35

Age	Age group
35	31-40
40	31-40
52	51-60
27	21-30
31	31-40
42	41-50
43	41-50
28	21-30
50	41-50
35	31-40

- One option is to group these ages into decades and create a categorical age variable.

Continuous Data - Tables

- We can then create a frequency table for this new categorical age variable

Interval	Freq.	Rel. Freq.
21-30	2	$2/10 = 0.2 = 20\%$
31-40	4	$4/10 = 0.4 = 40\%$
41-50	3	$3/10 = 0.3 = 30\%$
51-60	1	$1/10 = 0.1 = 10\%$
Total	10	$10/10 = 1.0 = 100\%$

- And create a bar chart to present this data graphically.

Arrangement

- Is the placement of visual elements within a display
- Can make a large difference in how easy it is to understand the data

Example: The following is a table accounting for produce deliveries over a weekend.

- What are the variables in this dataset?
- What object or event are we measuring?

	Friday	Saturday	Sunday
Morning	15	158	10
Afternoon	2	90	20
Evening	55	12	45

Handling Messy Data

We're measuring individual deliveries; the variables are Time, Day, Number of Products.

	Friday	Saturday	Sunday
Morning	15	158	10
Afternoon	2	90	20
Evening	55	12	45

Problem:

- Each column header represents a single *value* rather than a *variable*.
- Row headers are "hiding" the Day variable.
- The values of the variable, "Number of Products delivered", is not obvious.

Handling Messy Data

Need to reorganize the data to make explicit the event we're observing, and the variables associated with this event.

Delivery	Time	Day	No. of products
1	Morning	Friday	15
2	Morning	Saturday	158
3	Morning	Sunday	10
4	Afternoon	Friday	2
5	Afternoon	Saturday	90
6	Afternoon	Sunday	20
7	Evening	Friday	55
8	Evening	Saturday	12
9	Evening	Sunday	45

Why Is Data Dirty?

- Incomplete data may come from
  - “Not applicable” data value when collected
  - Different considerations between the time when the data was collected and when it is analyzed.
  - Human/hardware/software problems
- Noisy data (incorrect values) may come from
  - Faulty data collection instruments
  - Human or computer error at data entry
  - Errors in data transmission
- Inconsistent data may come from
  - Different data sources
  - Functional dependency violation (e.g., modify some linked data)
- Duplicate records also need data cleaning

Duplicate Data

Removing duplicates:

- You have a database of customer information and
- you find that there are duplicate records for the same customer,
- you can remove the duplicates to ensure that your data is accurate and up-to-date.

Merging duplicates:

- You have a database of product information and
- you find that there are duplicate records for the same product,
- you can merge the duplicates into a single record to ensure that your data is complete and consistent.

Flagging duplicates:

- If you have a database of employee information and
- you find that there are duplicate records for the same employee,
- you can flag the duplicates so that they can be easily identified and removed or merged.

Data Cleaning and Preparation

Data Inspection

- Examine the dataset to understand its structure, format, and potential issues. Check for the number of rows, columns, data types, and any missing values.
- Significant amount of time is spent on data preparation:
  - take up 80% or more of an analyst's time

Data Handling

- Duplicates: Identify and handle duplicate entries to avoid bias and inconsistencies in the analysis.
- Outliers: Decide whether to remove, transform, or impute outliers depending on the context and impact on the analysis.
- Missing Data: Missing data occurs commonly in many data analysis applications.

Outliers

- Outliers are data points that significantly deviate from the majority of the data and may arise due to:
  - measurement errors, data entry mistakes or rare events.
- Approaches to handle outliers:
  - **Identify Outliers:** Using the z-score or the interquartile range (IQR)
    - Z-score measures how many standard deviations a data point is from the mean.
    - Any data point with an absolute Z-score greater than 3 is considered an outlier.
  - **Transform Data:** Instead of removing outliers, you can try transforming the data
    - using log transformation, square root, or Box-Cox transformation.
    - may help make the data more normally distributed and reduce the impact of extreme values.
- Choice of outlier handling technique should be based on the characteristics of the data and the specific requirements of the analysis or modelling task.

Handling Missing Values

Deletion

- Remove entire rows with missing values. Simple but may lead to data loss, especially if the missing values are not randomly distributed.
- **Dropping Columns:** If a column contains a large number of missing values and is not essential for the analysis, it may be dropped from the dataset.

Imputation

- Mean/Median/Mode Imputation
- Interpolation: Use the values of neighbouring data points to estimate missing values, especially in time-series or sequential data.
- Regression Imputation: Predict missing values using regression models based on other variables with complete data.
- K-Nearest Neighbours (KNN) Imputation: Estimate missing values based on the average of the K-nearest data points in the feature space.
- Multiple Imputation: Generate multiple imputed datasets and combine the results to account for uncertainty in imputing missing value

Categorical Imputation:

- Assign a new category or label to missing categorical data.
- Use the most frequent category (mode) to impute missing categorical values.

Handling Missing Values

Mean Imputation

- Replace missing values with the mean of the non-missing values.
- It is appropriate when the data has a normal distribution, and the missing values are missing at random.

Median Imputation

- Replace missing values with the median of the non-missing values in the column.
- Median imputation is robust to outliers and can be a better choice when dealing with skewed data.

Mode Imputation

- Replace missing categorical values with the mode (most frequent value) of the non-missing values in that column.
- Used for categorical data and non-numeric variables

Take-home message

- Data inspection is the first and important step in EDA
- Pay attention to methods used in data cleaning
- Deleting is not always the best option
- The way the data is represented needs to be useful for what we want to find out about it.